CAT403 COMPUTING INFRASTRUCTURE MAJOR PROJECT

PROJECT PROGRESS DEMO



COGNISHIELD

A Large-Language Driven Campus Network Automation

Brought to You By:

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WHAT DOES IT DO?



COGNISHIELD

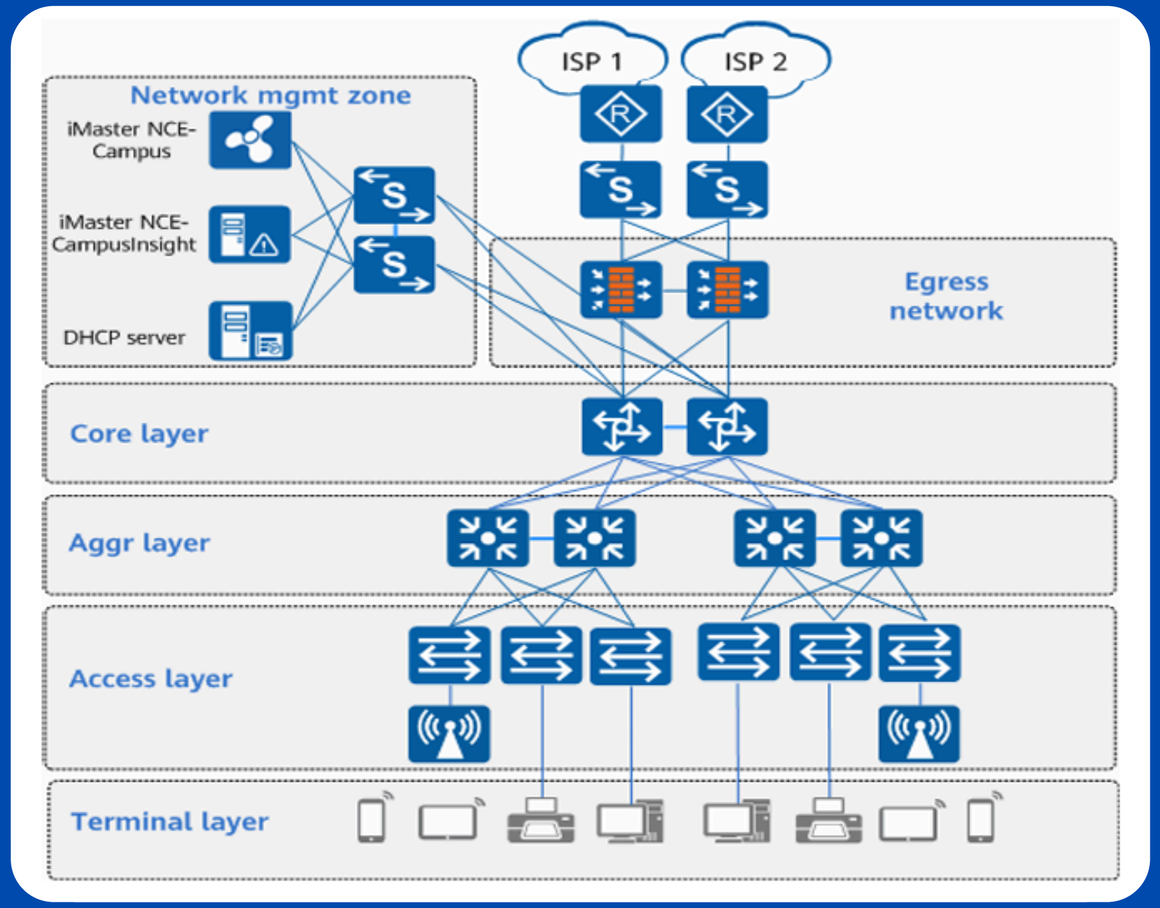
A Large-Language Driven Campus Network Automation

COGNISHIELD

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: Here are some things you should know first before you know more about me :)

WHAT IS **CAMPUS NETWORK?**



A proprietary local area network(LAN) or a group of connected local networks that belong to one organization.

It allows different buildings within the same area to communicate and share resources



UNIVERSITIES

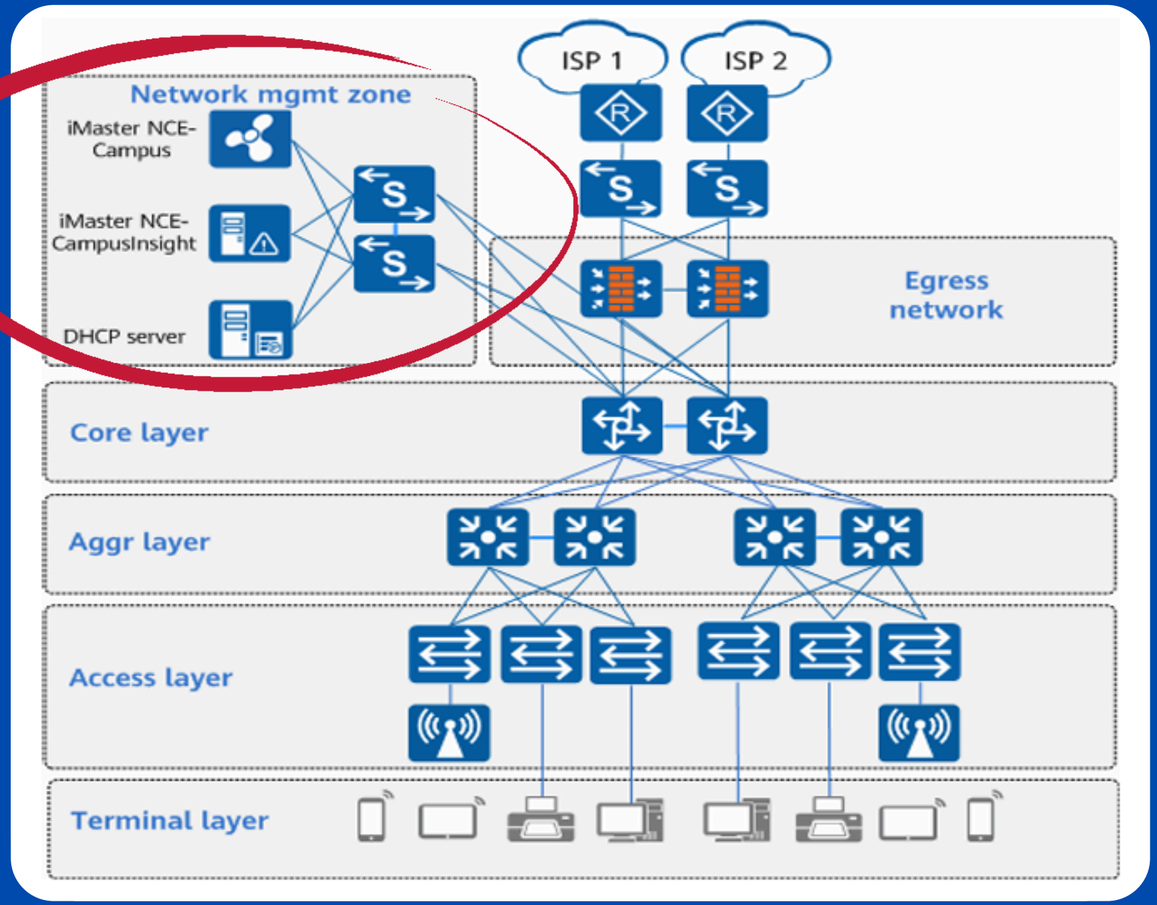


OFFICES



HOSPITALS

NETWORK MANAGEMENT ZONE

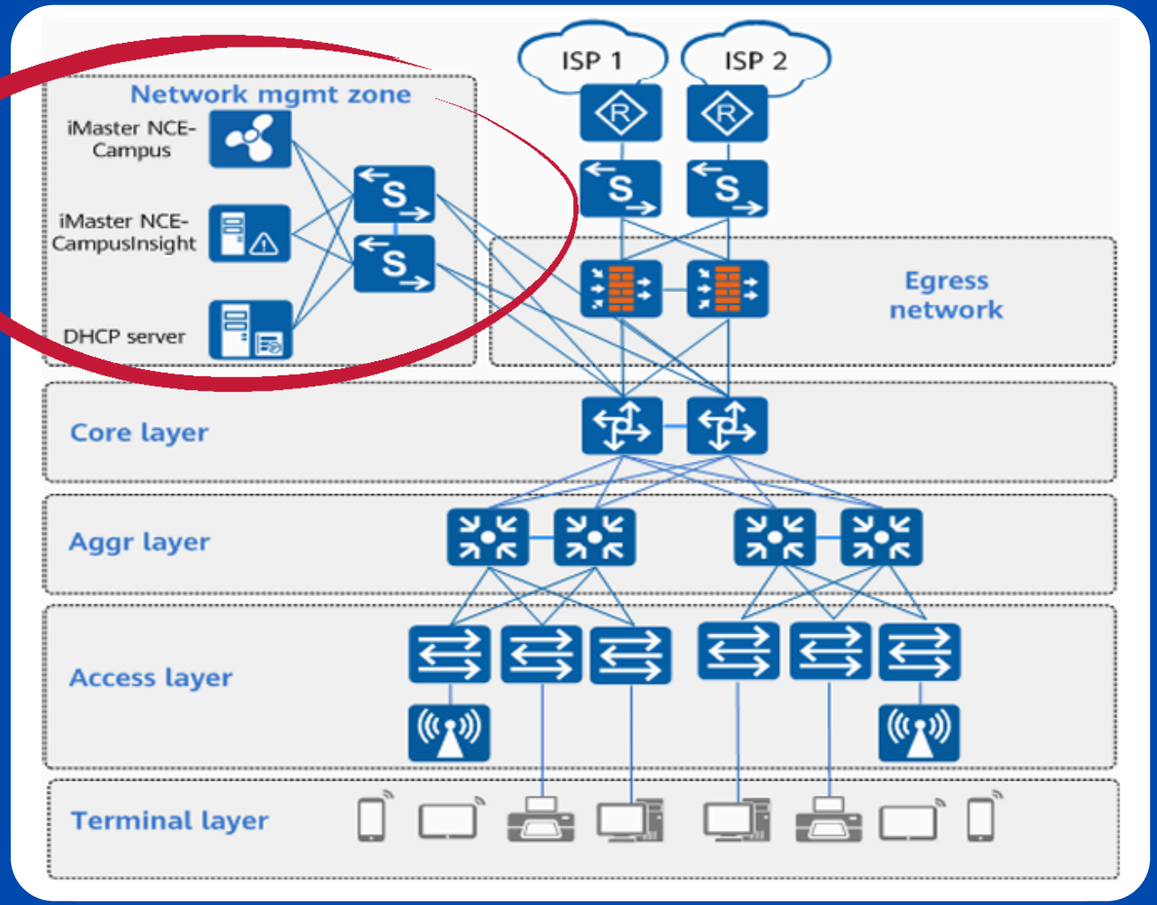


Dedicated part of a network used specifically to monitor, control, and manage network devices and services.

Why It Exists?

- Complex and growing network architectures
- Time-consuming manual configuration
- High risk of errors and misconfigurations
- Limited real-time visibility and slow issue response

NETWORK MANAGEMENT ZONE



Dedicated part of a network used specifically to monitor, control, and manage network devices and services.



NETWORK MANAGEMENT SERVER



LOGGING & MONITORING SYSTEMS



CONFIGURATION & AUTOMATION SERVERS



NETWORK MANAGEMENT SERVER



LOGGING & MONITORING SYSTEMS



CONFIGURATION & AUTOMATION SERVERS

Integrated Network Management & Automation Platform

DNA Center

JUNIPER driven by Mist AI

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Analytics

Management Control

iMaster NCE

QUESTION:

“So, if we already have all these existing platforms from **Cisco**, **Huawei**, **Juniper**, etc, why do we still need **CogniShield**?”



PROBLEM STATEMENT

- **Proprietary & costly platforms**
- **No open, vendor-neutral alternatives**
- **Lack of LLM-driven automation**







BACKGROUND STUDY

COMPETITOR ANALYSIS

Feature / Capability	CogniShield	Cisco DNA Center	Juniper Mist	Huawei iMaster NCE	NetBrain
Open-Source					
Vendor-Neutral					
LLM Intent Automation					
Intent-Based Automation					
Closed-Loop Remediation					
Deployment / Hardware					
Cost & Accessibility					
Education / Research					

What will be the EFFECTS?

THE EFFECTS

1 Proprietary systems widen the technology gap

Vendor lock-in causes only the enterprise can access to the technology.

Normal communities like students, researchers, startups have no access to the technology.

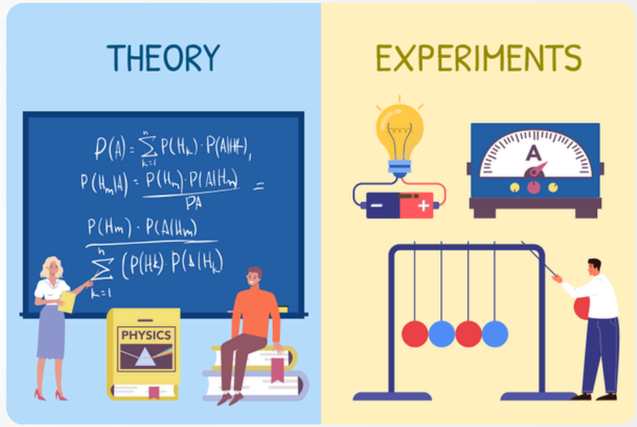


2 Beyond Affordable by SMEs

- Manual, script-heavy network operations
- High overhead, greater risk of errors
- Slow & less-scalable networks

3 Limited Exposure to Students:

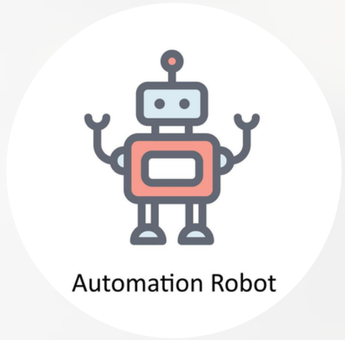
- Stick to theory, CLI-lab
- Skill gap between academy & industry
- Slow down innovation, less community-driven



SYSTEM OBJECTIVES







To Develop an Open & Vendor-Neutral Network Automation Platform

To Implement LLM-driven, Intent-based Configuration As Code (CaaC) approach

To Enable Automated Incident Response with Human Oversight



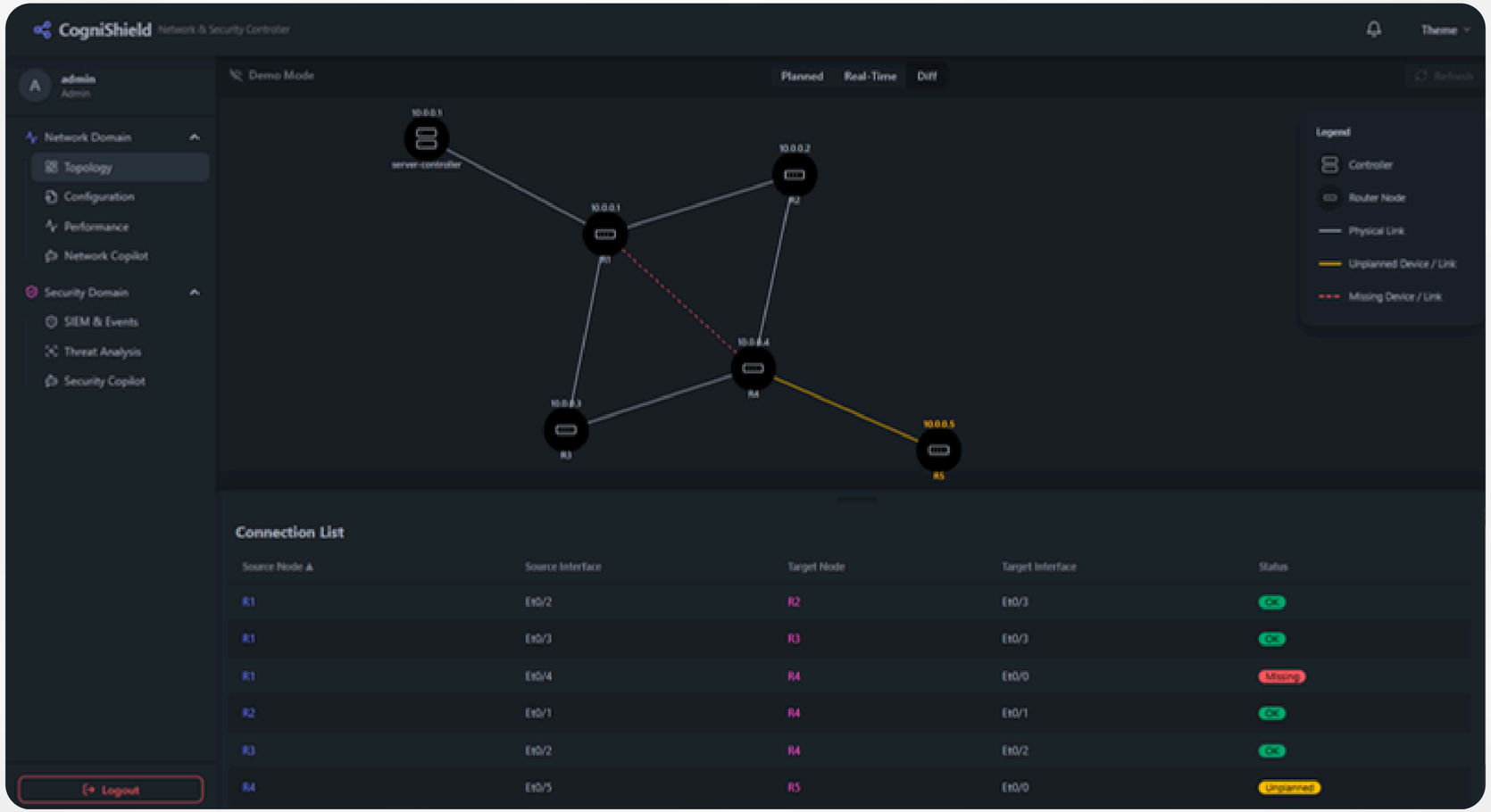
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“CogniShield is a free, open-source intelligent network management system that understands operator intent, automates network configurations, continuously monitors the network, detects and fixes issues, and alerts operators in real time.”

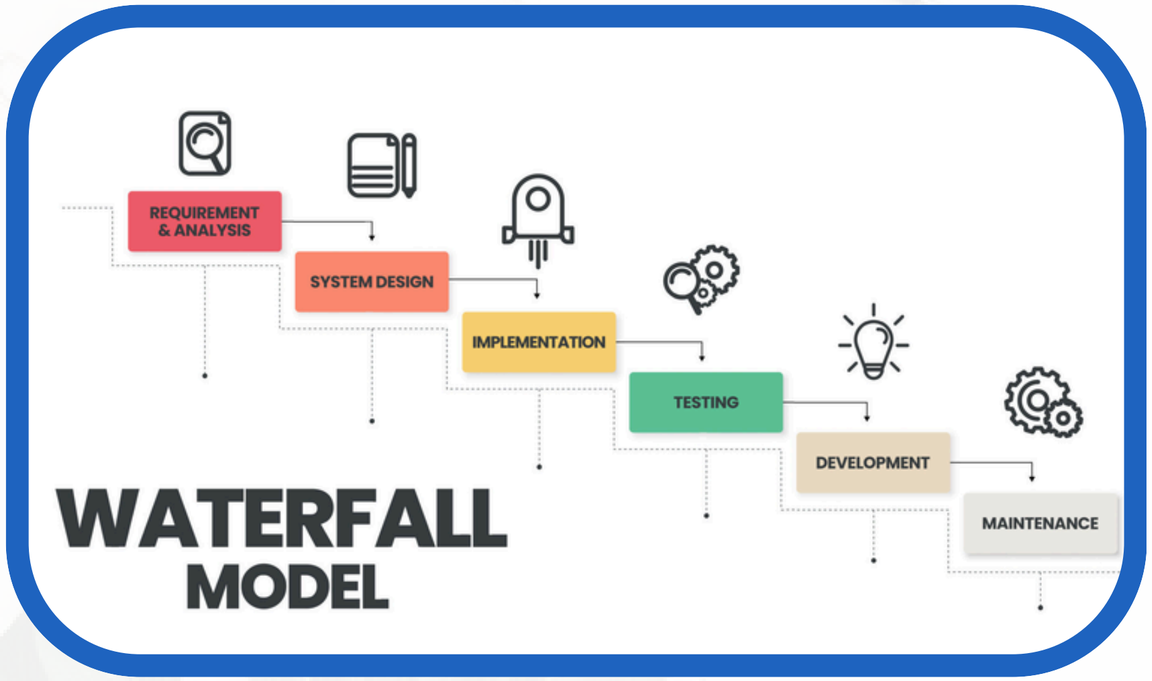
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DEVELOPMENT METHODOLOGY

DEVELOPMENT METHODOLOGY



WATERFALL MODEL

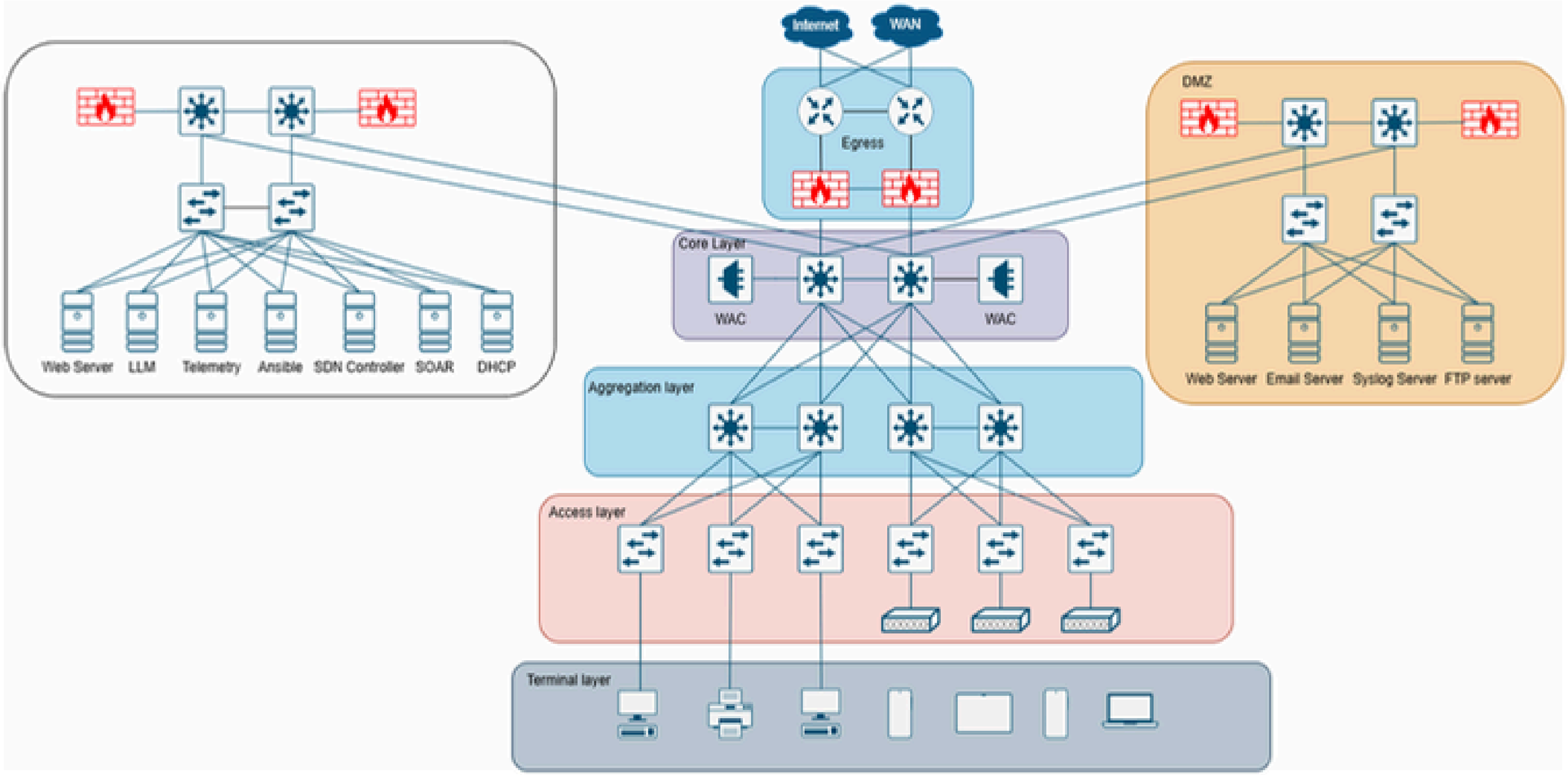
Structured & sequential development

Strong documentation & validation

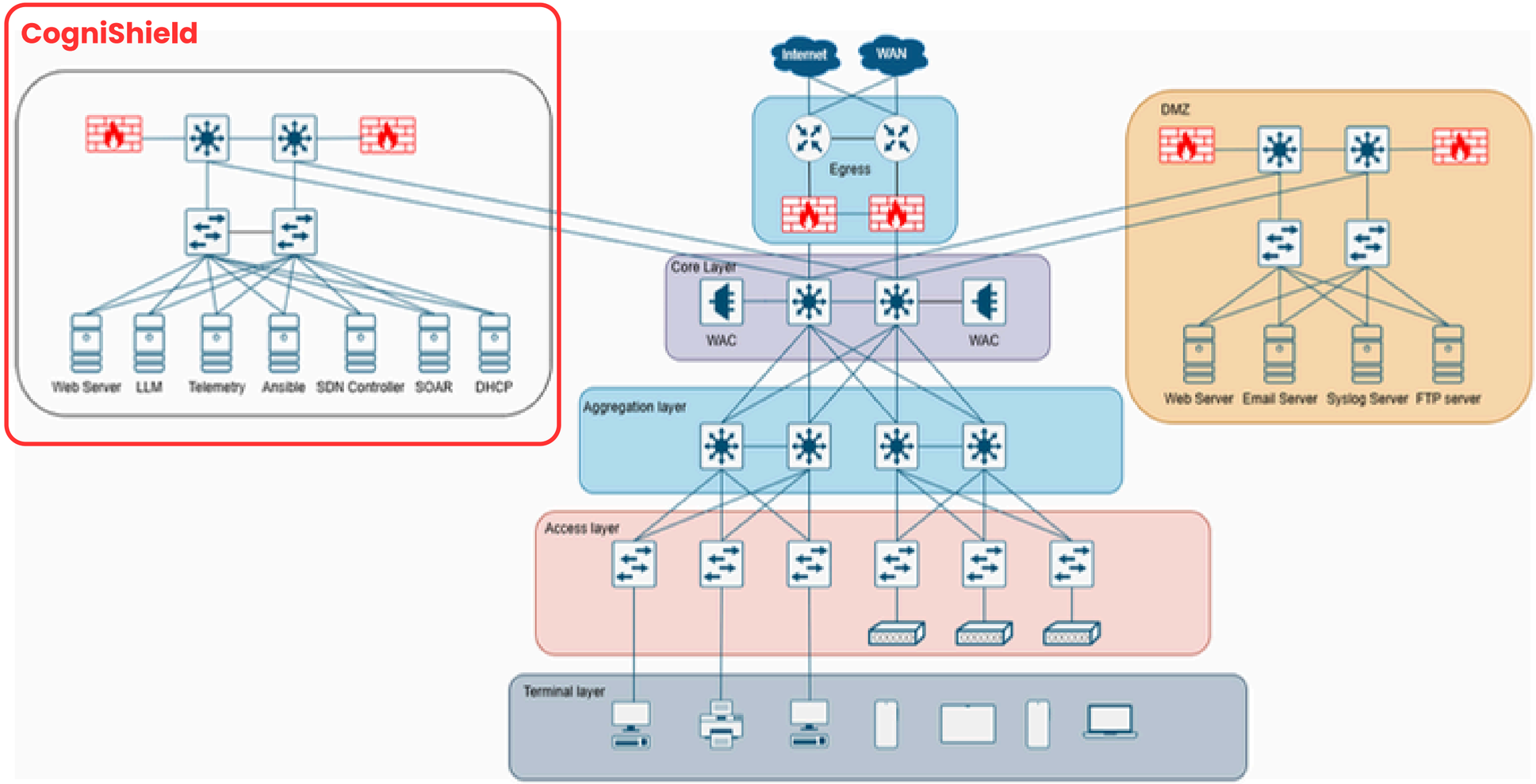
Suitable for safety-critical network systems

SYSTEM DESIGN

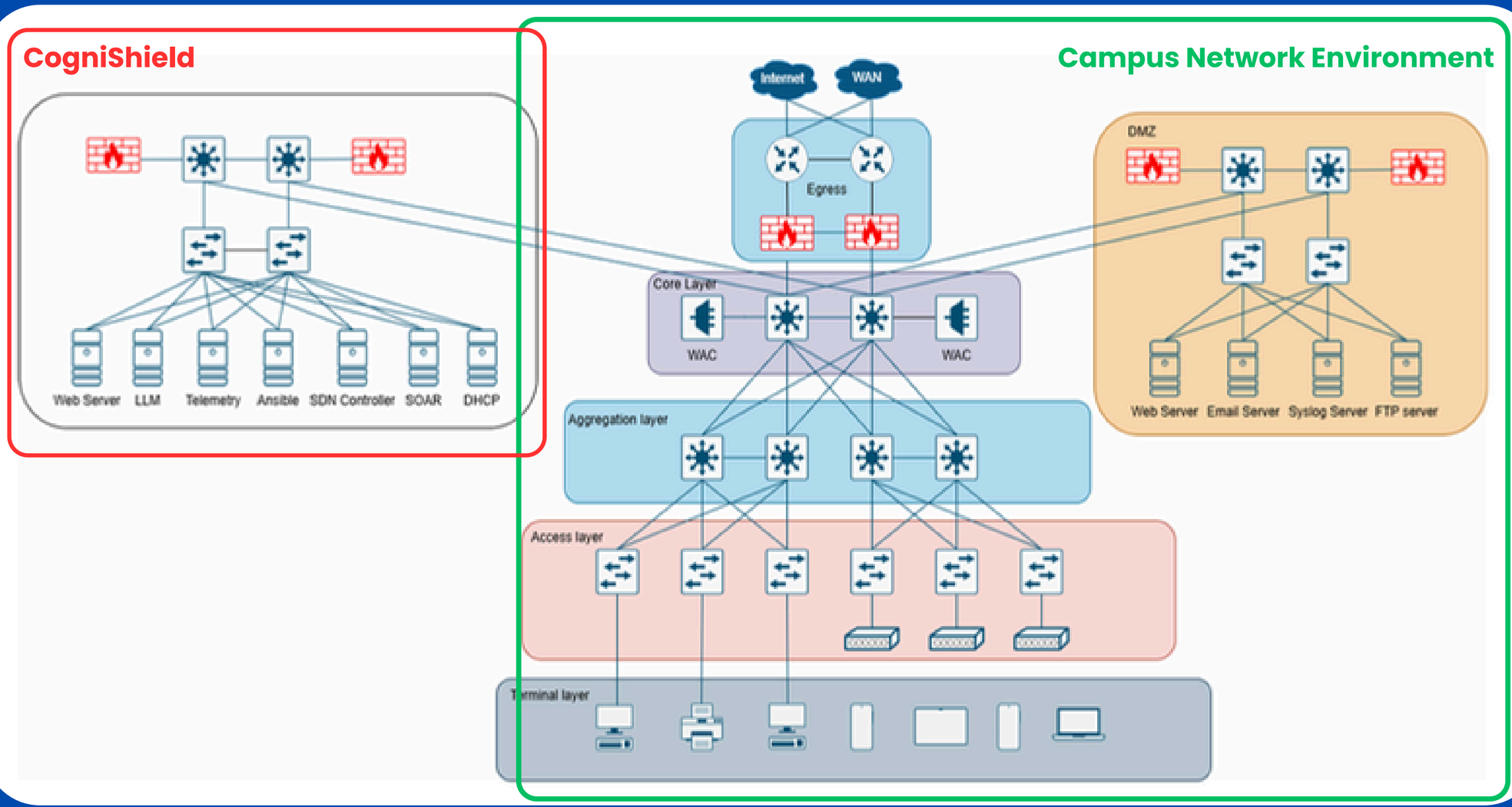
HIGH-LEVEL SYSTEM ARCHITECTURE



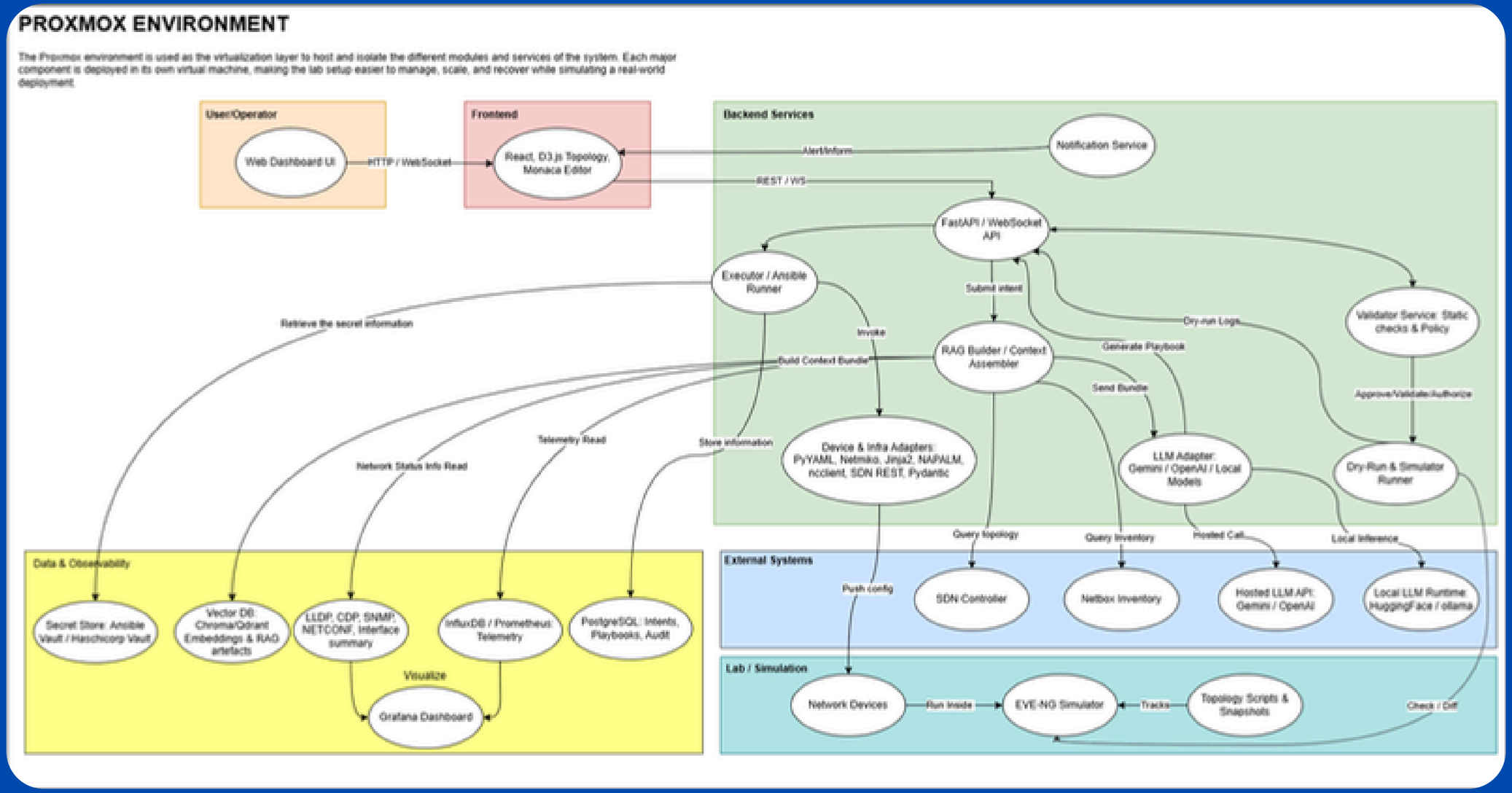
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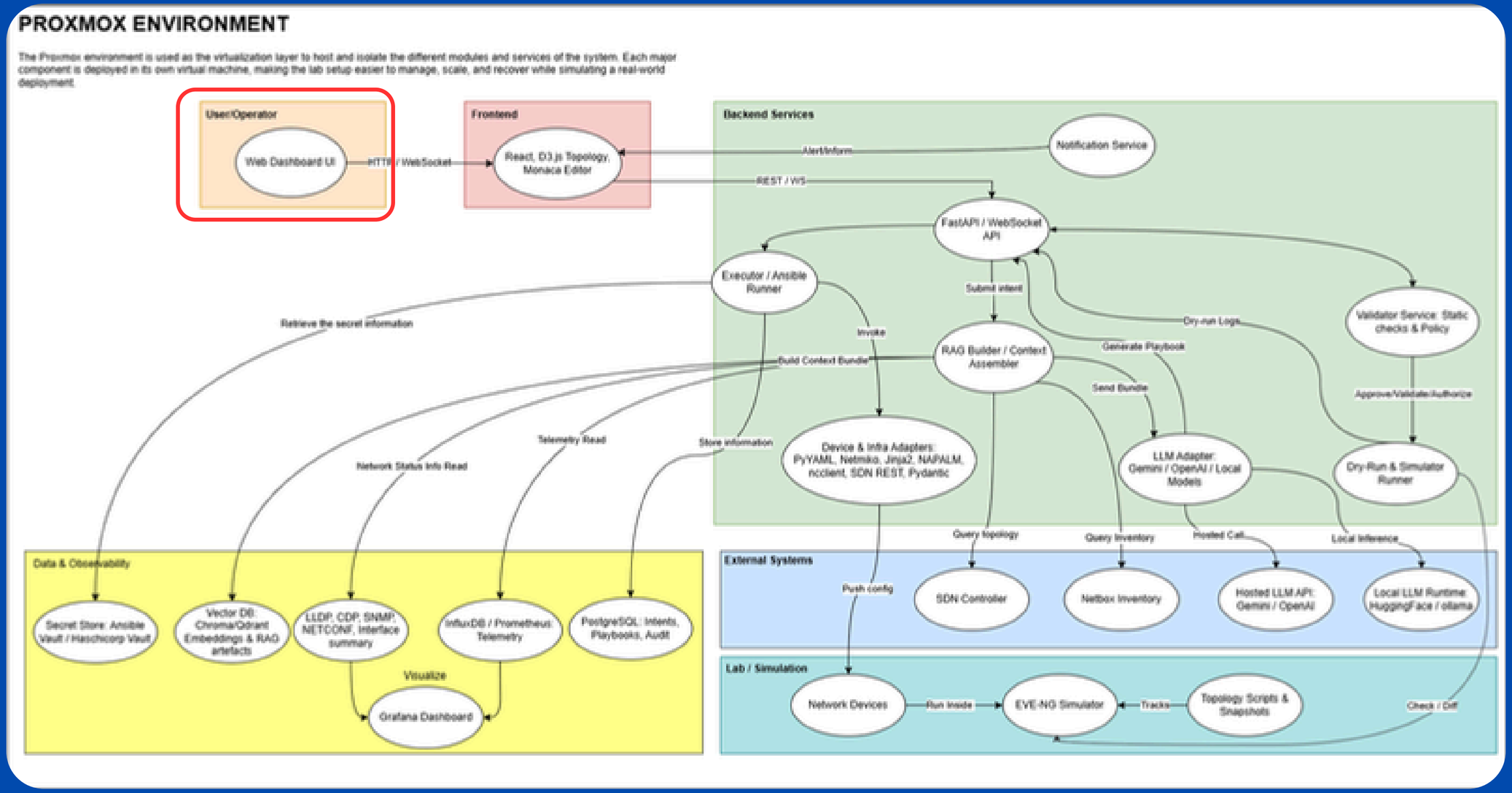
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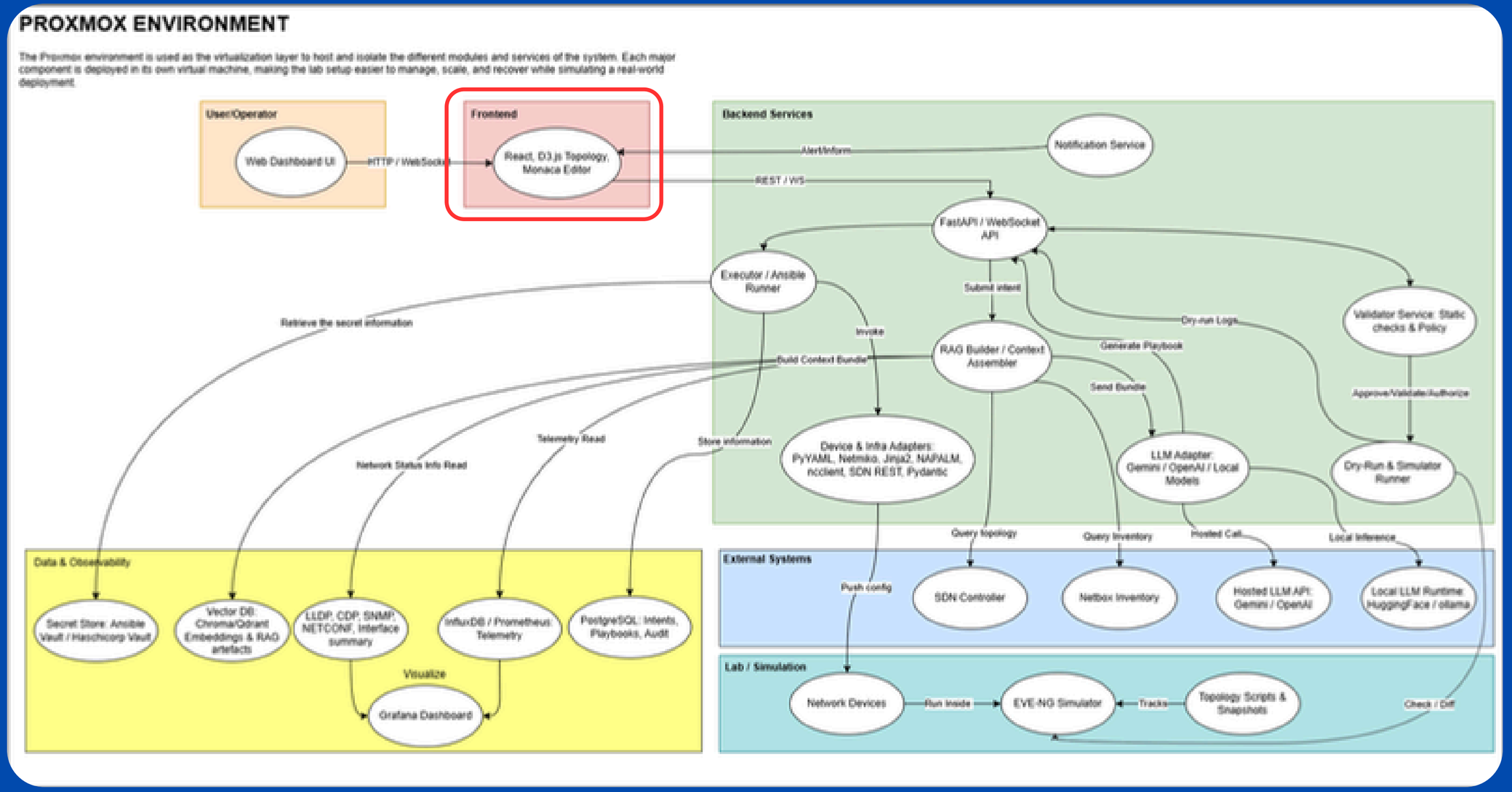
LOW-LEVEL SYSTEM ARCHITECTURE



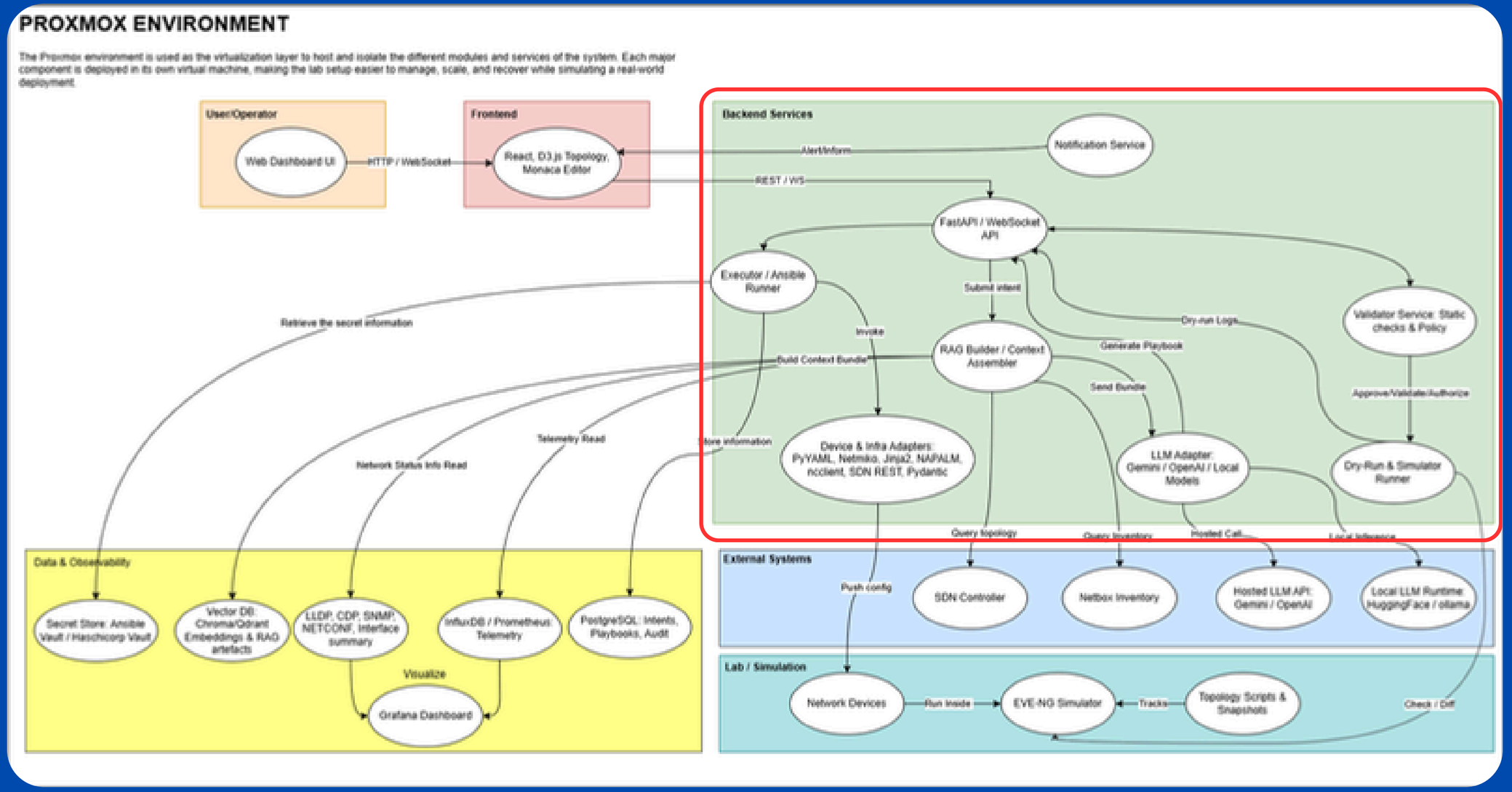
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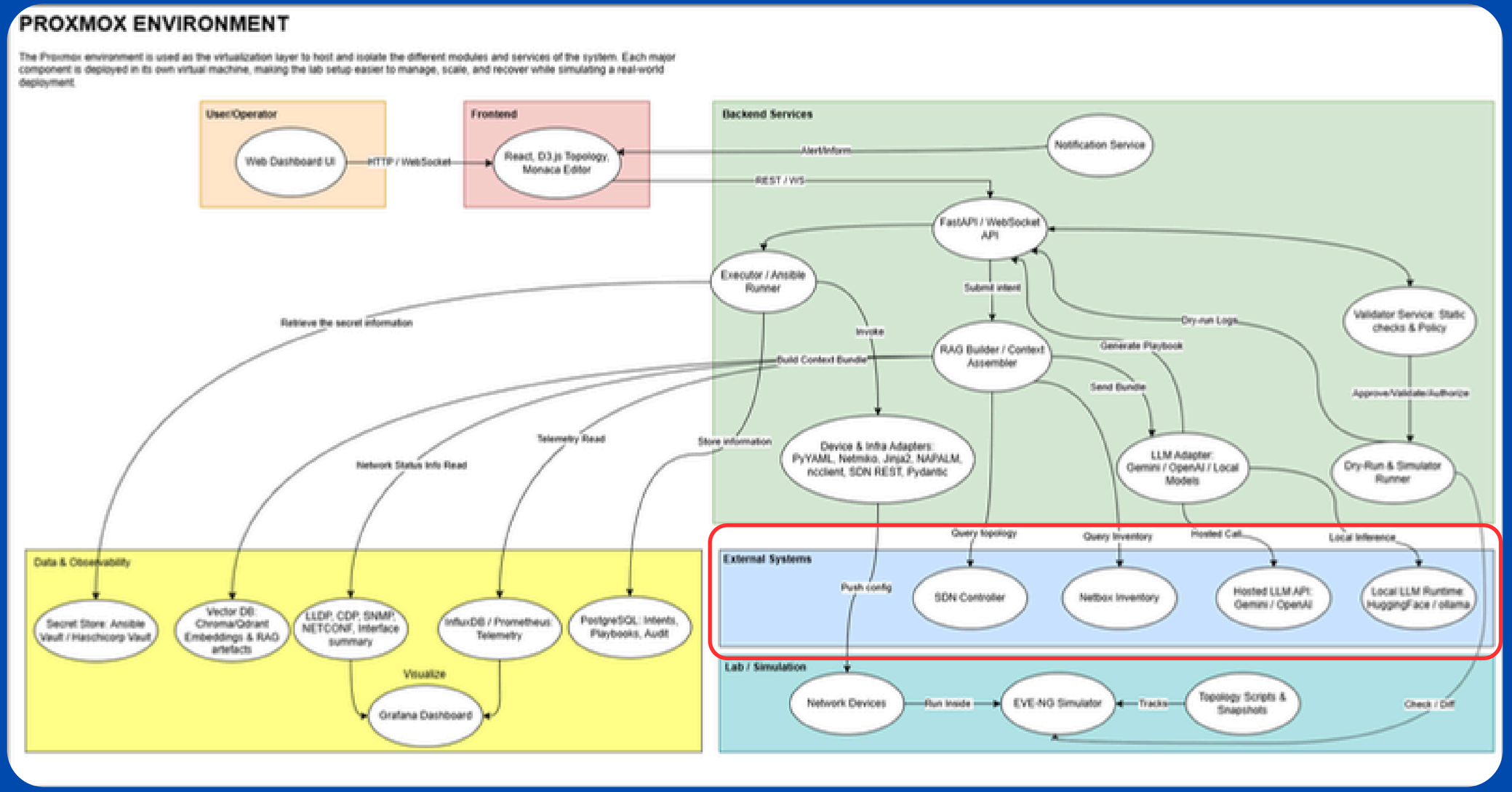
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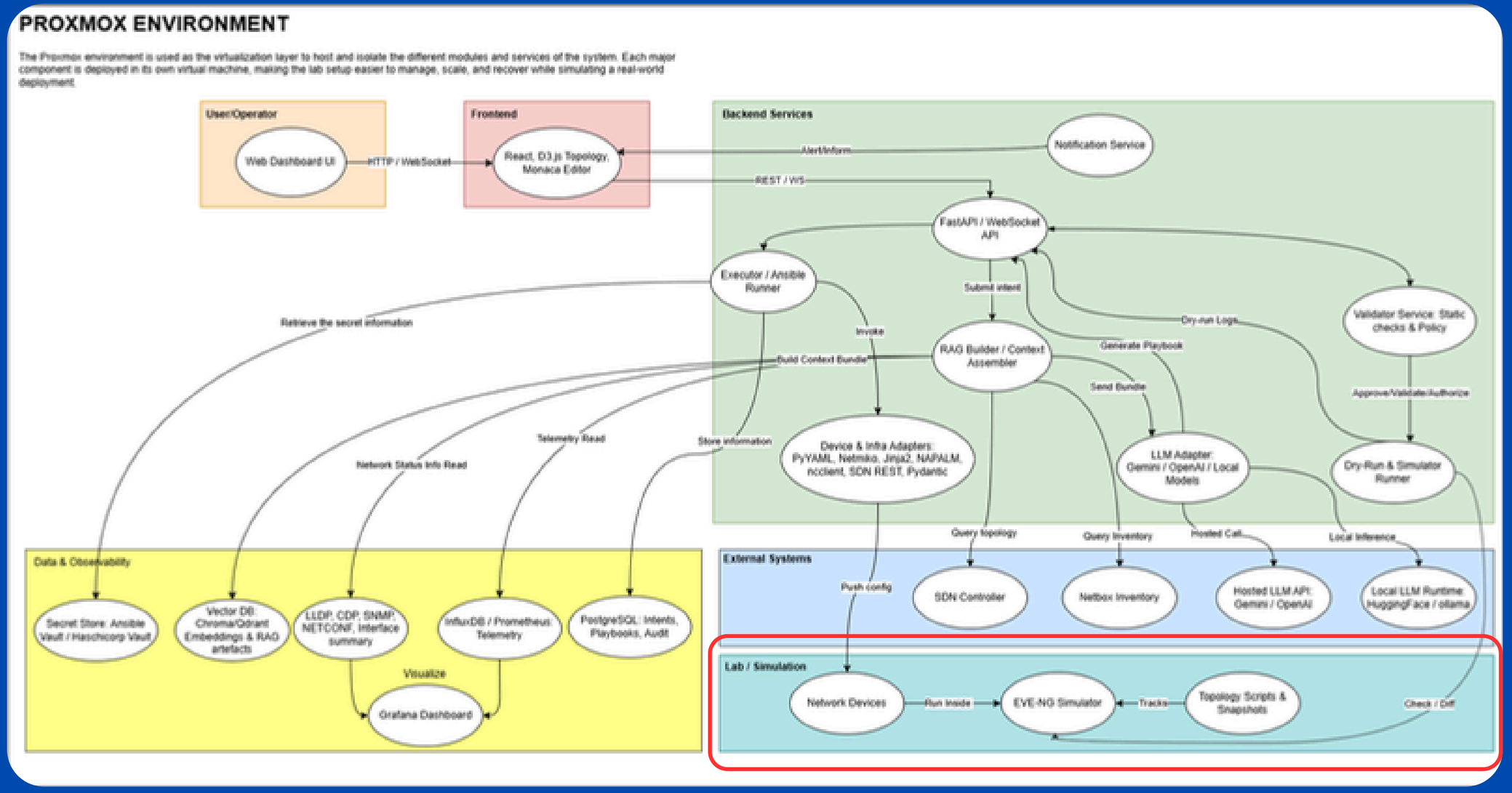
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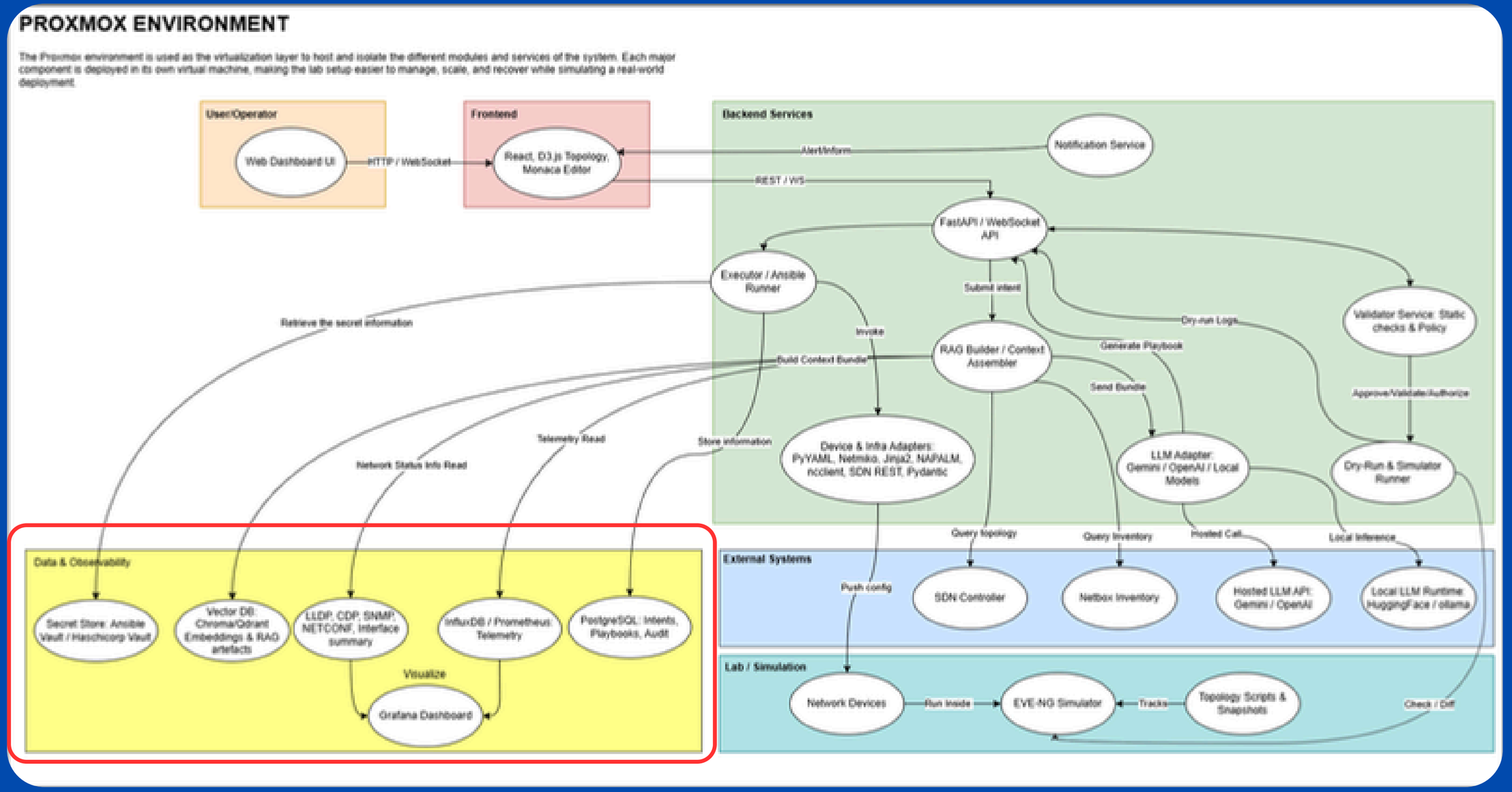
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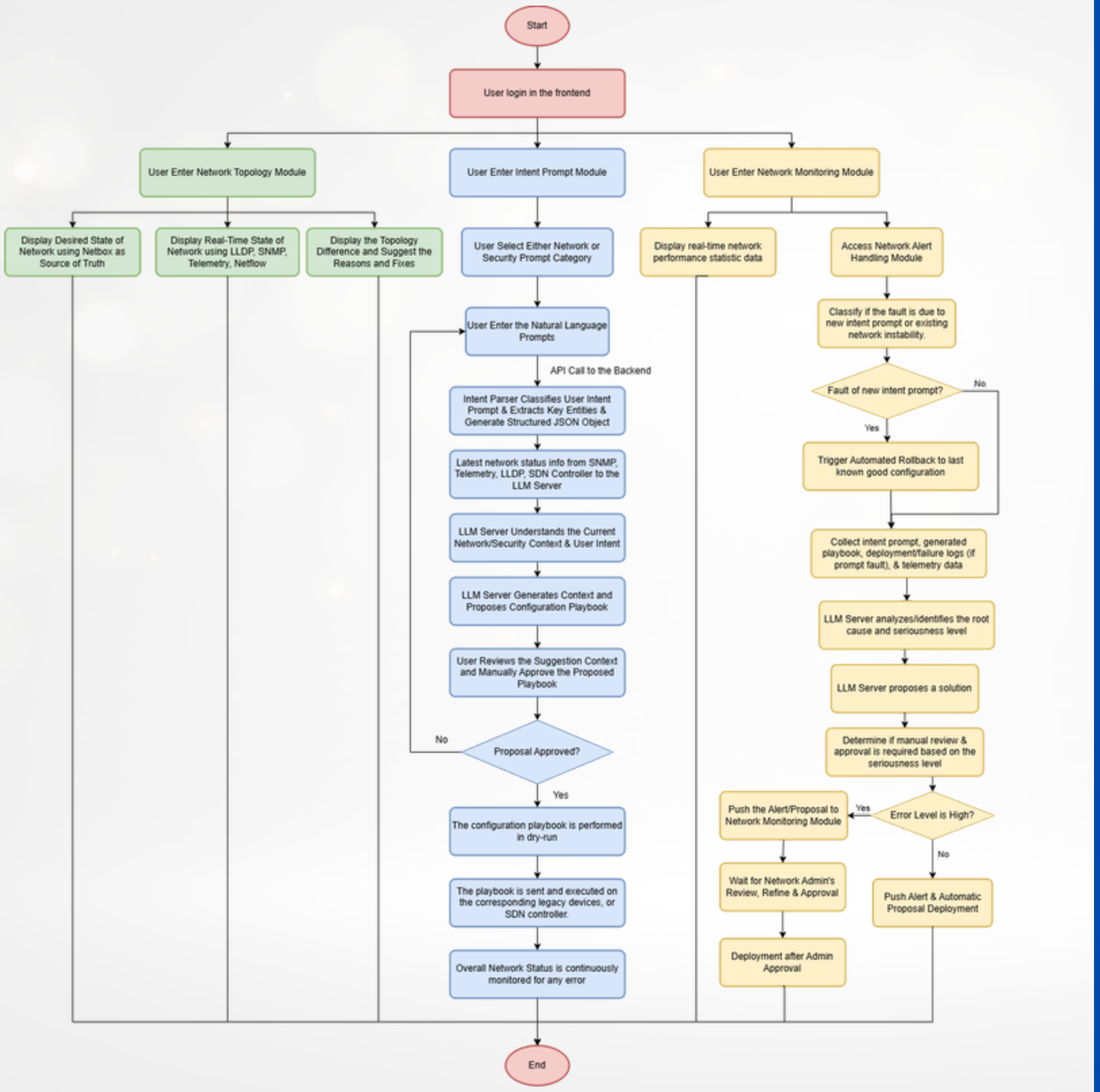


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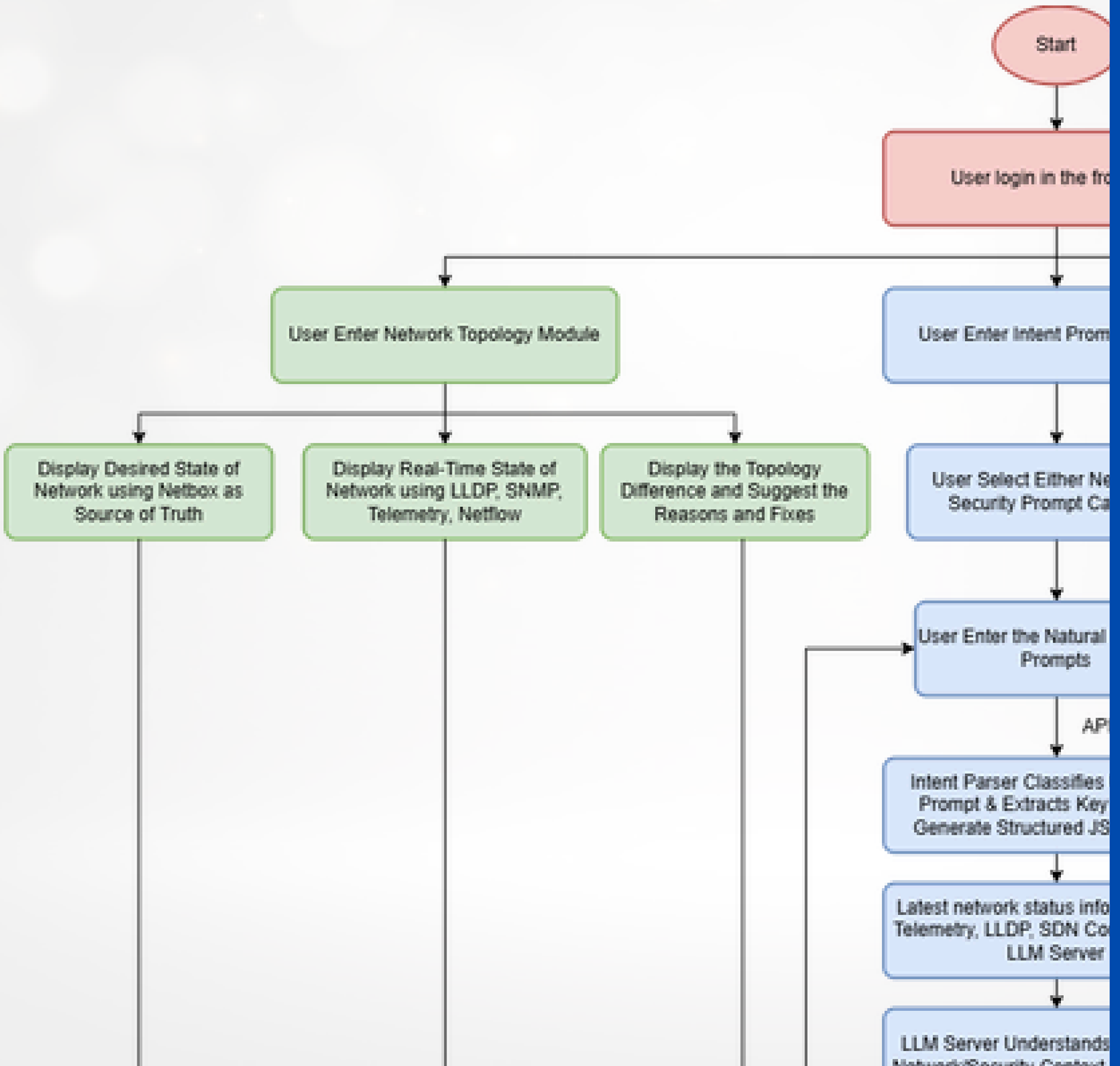
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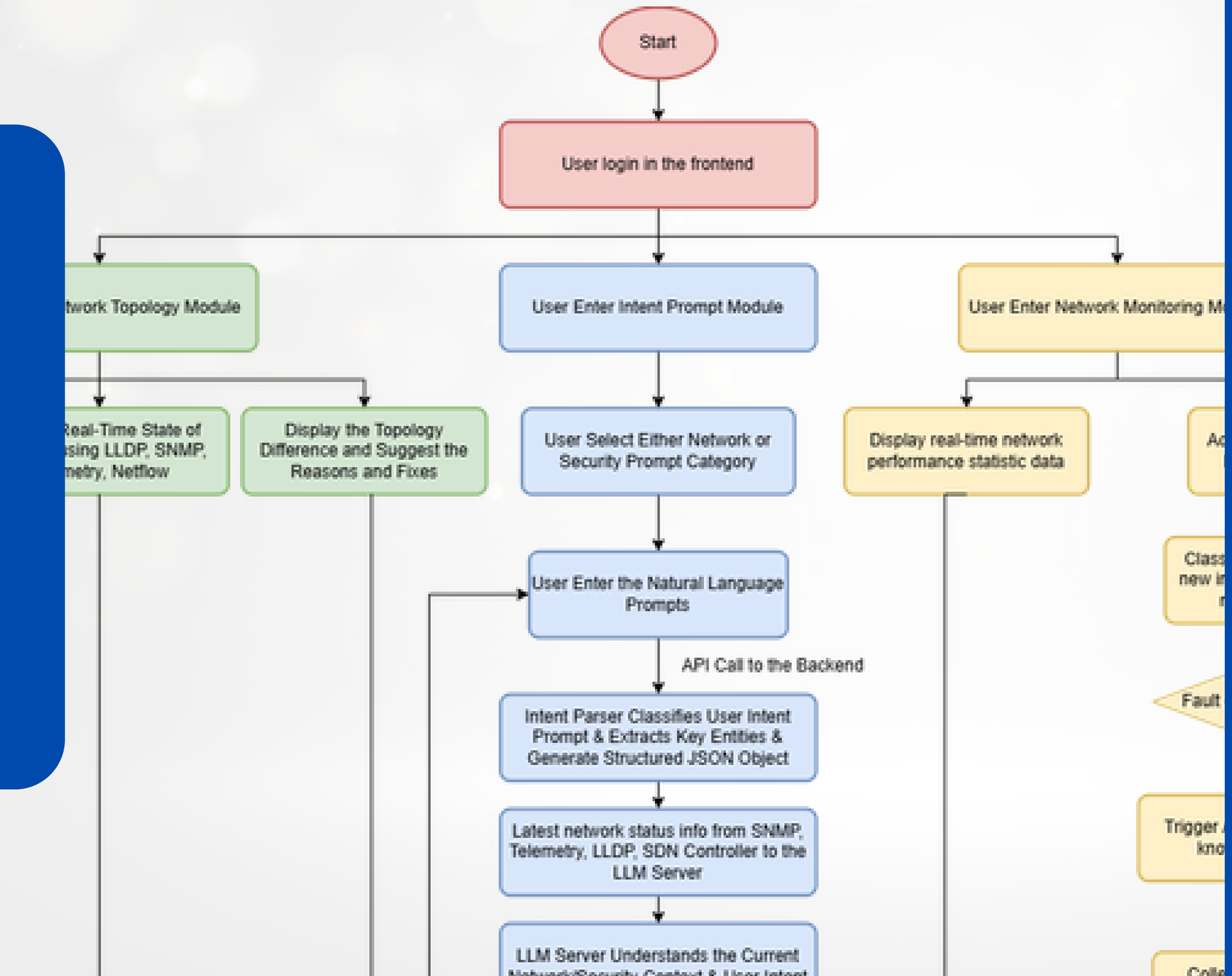




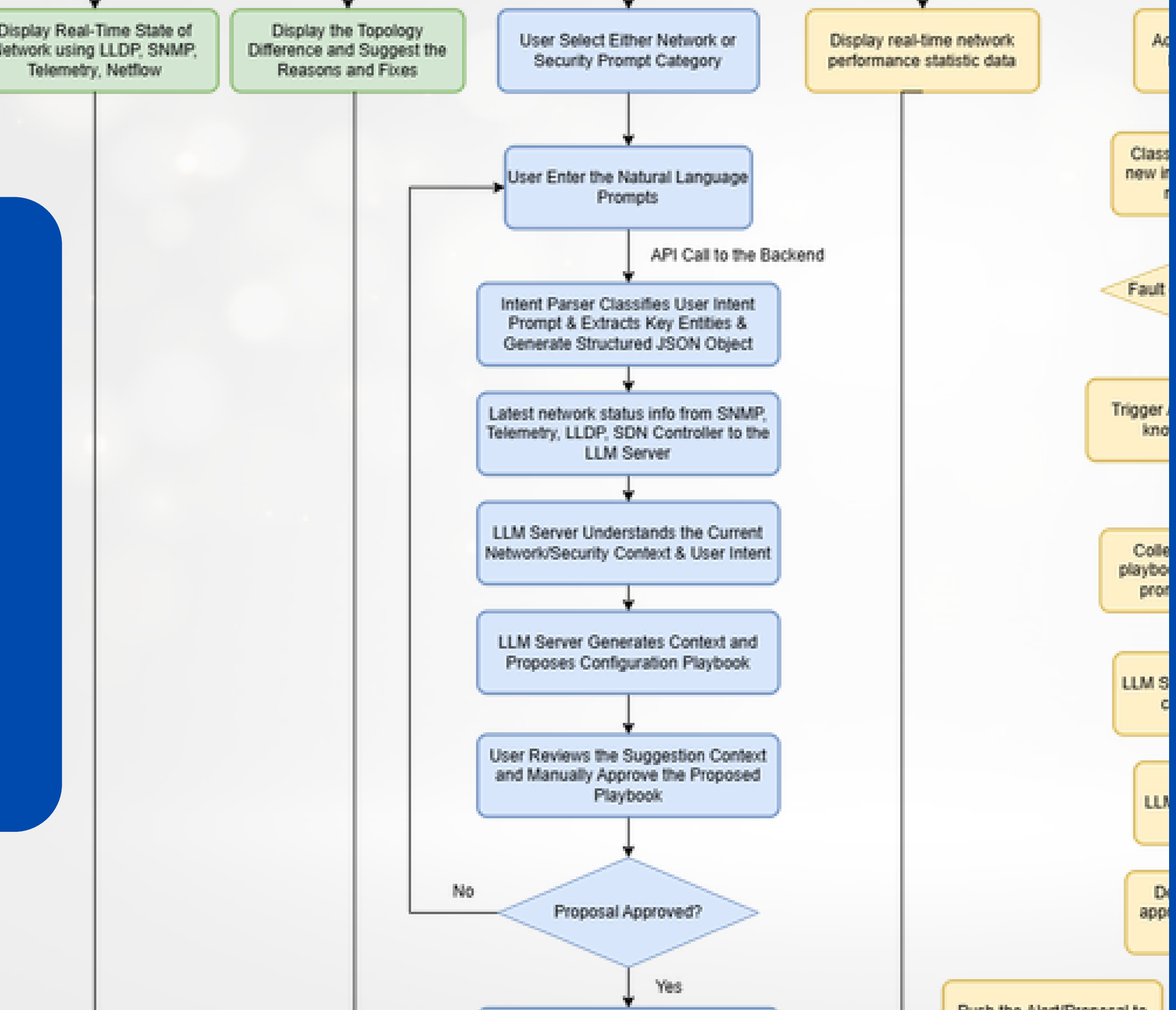
USER ACTIVITY DIAGRAM

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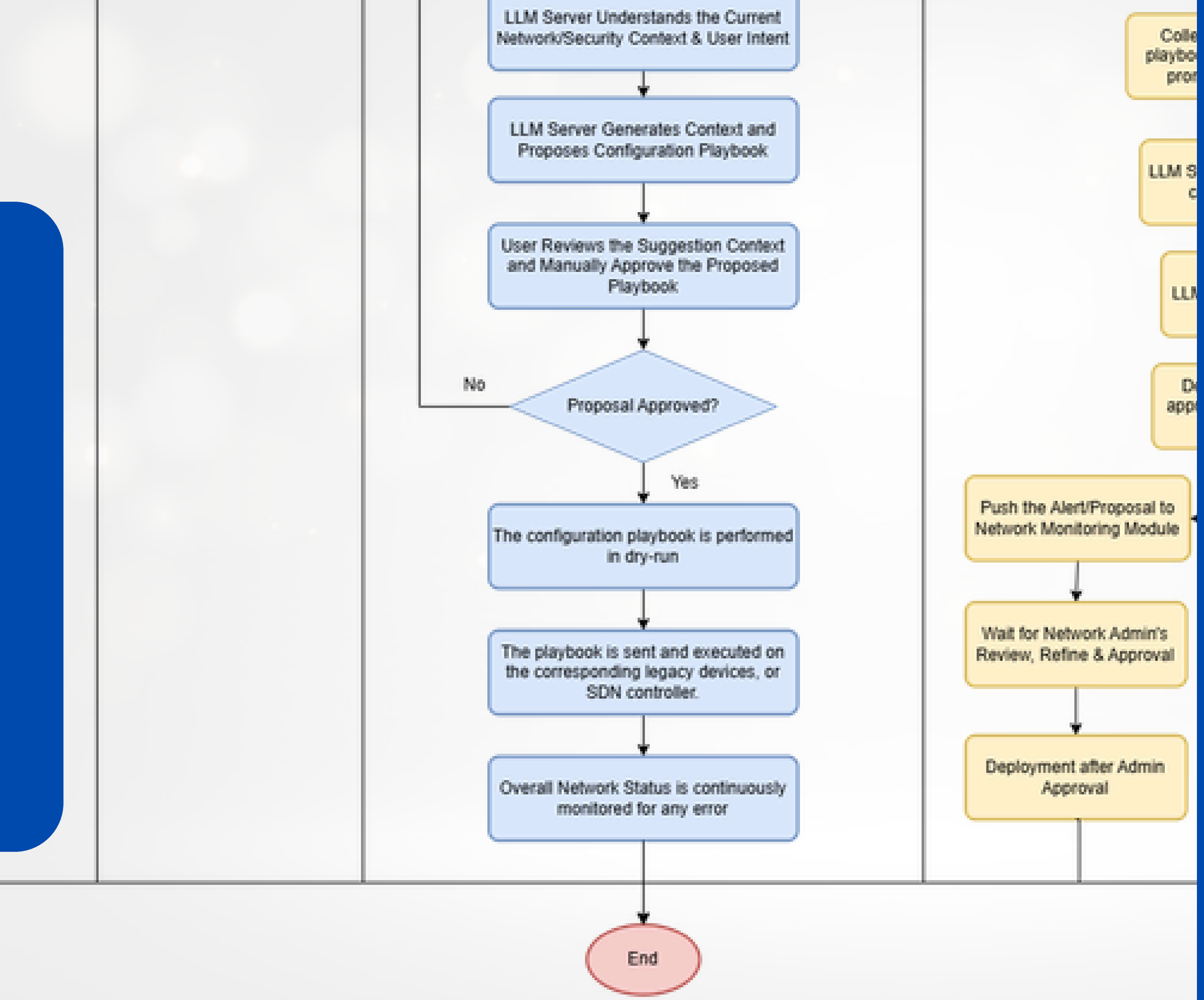




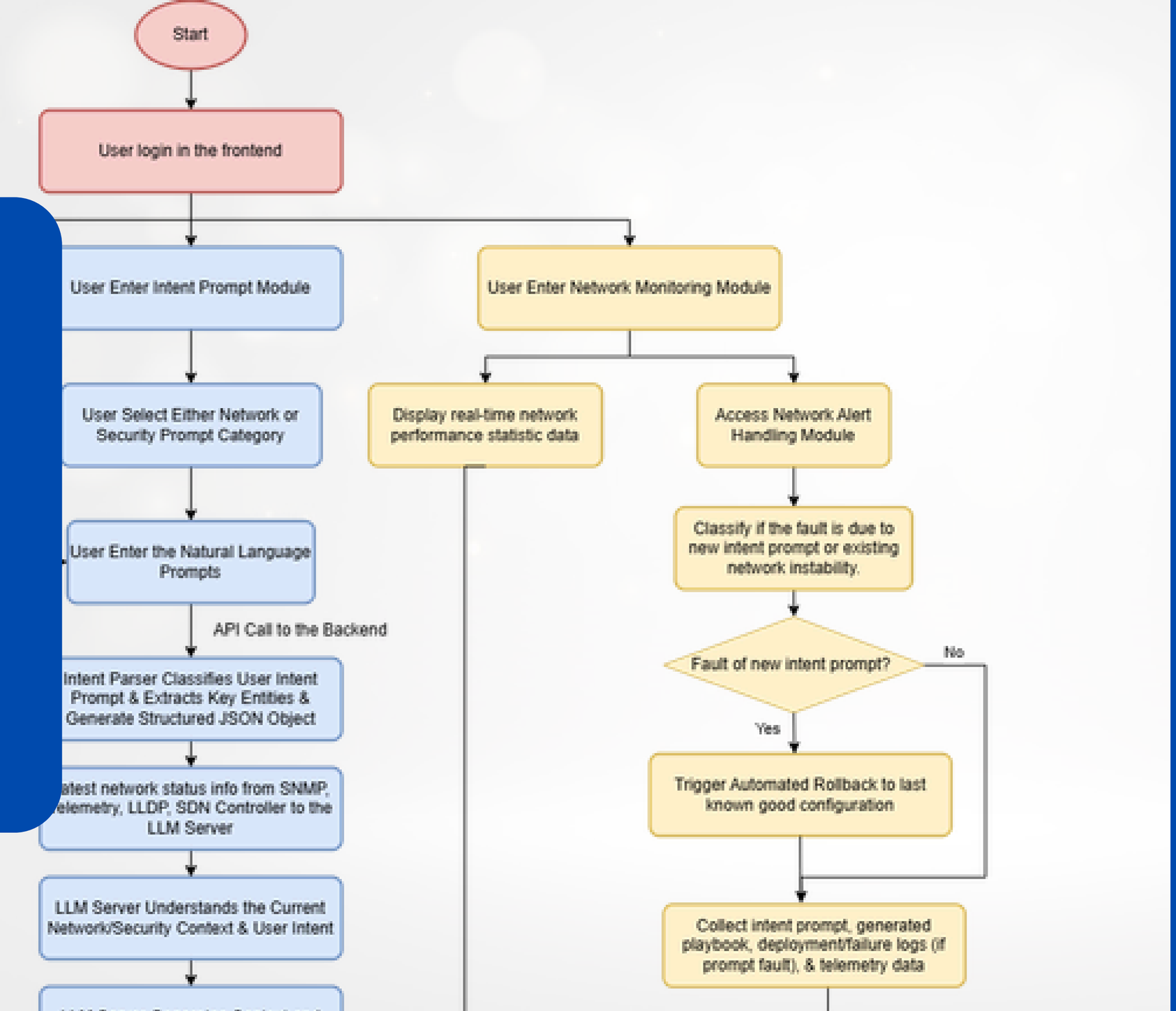
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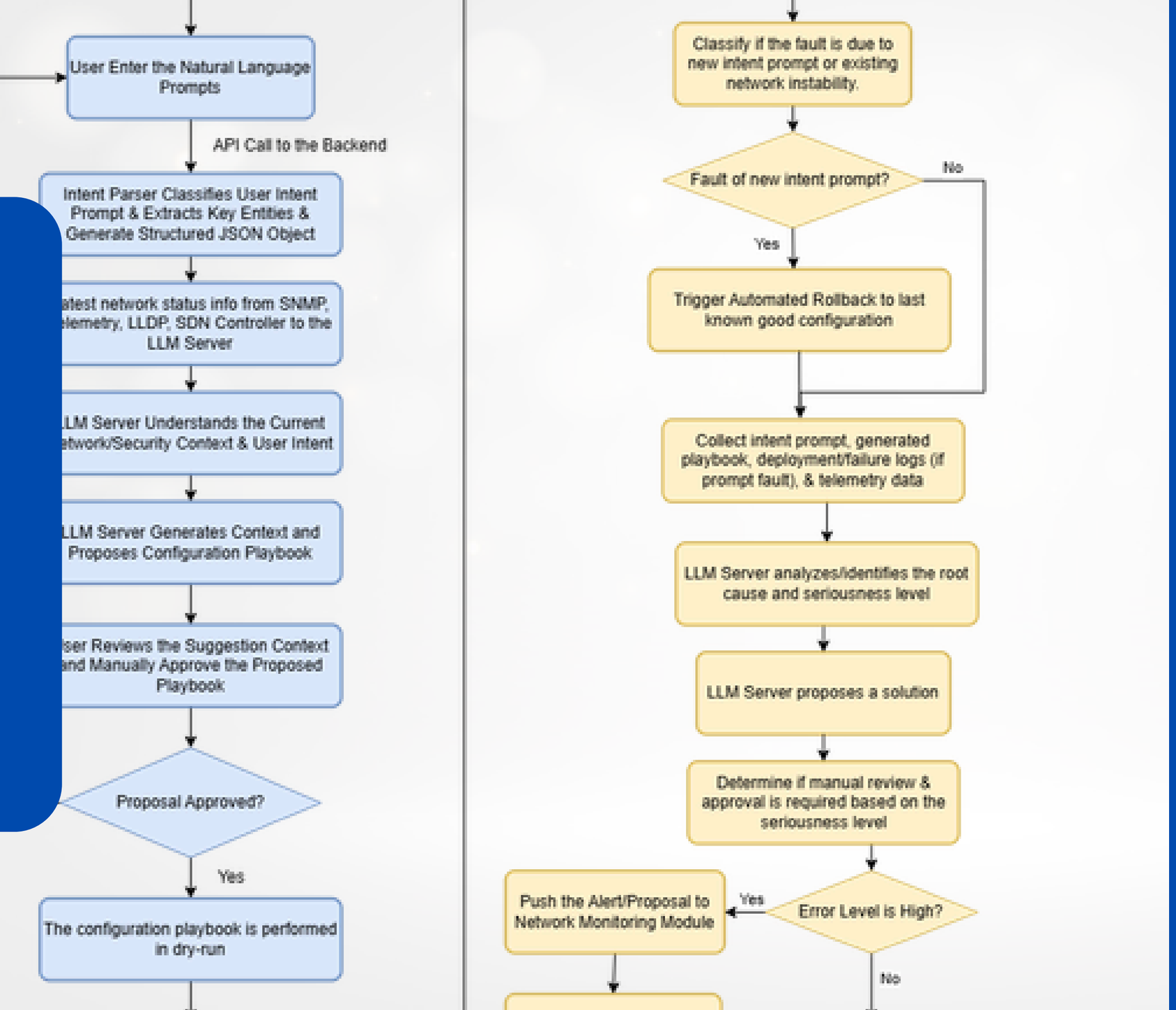
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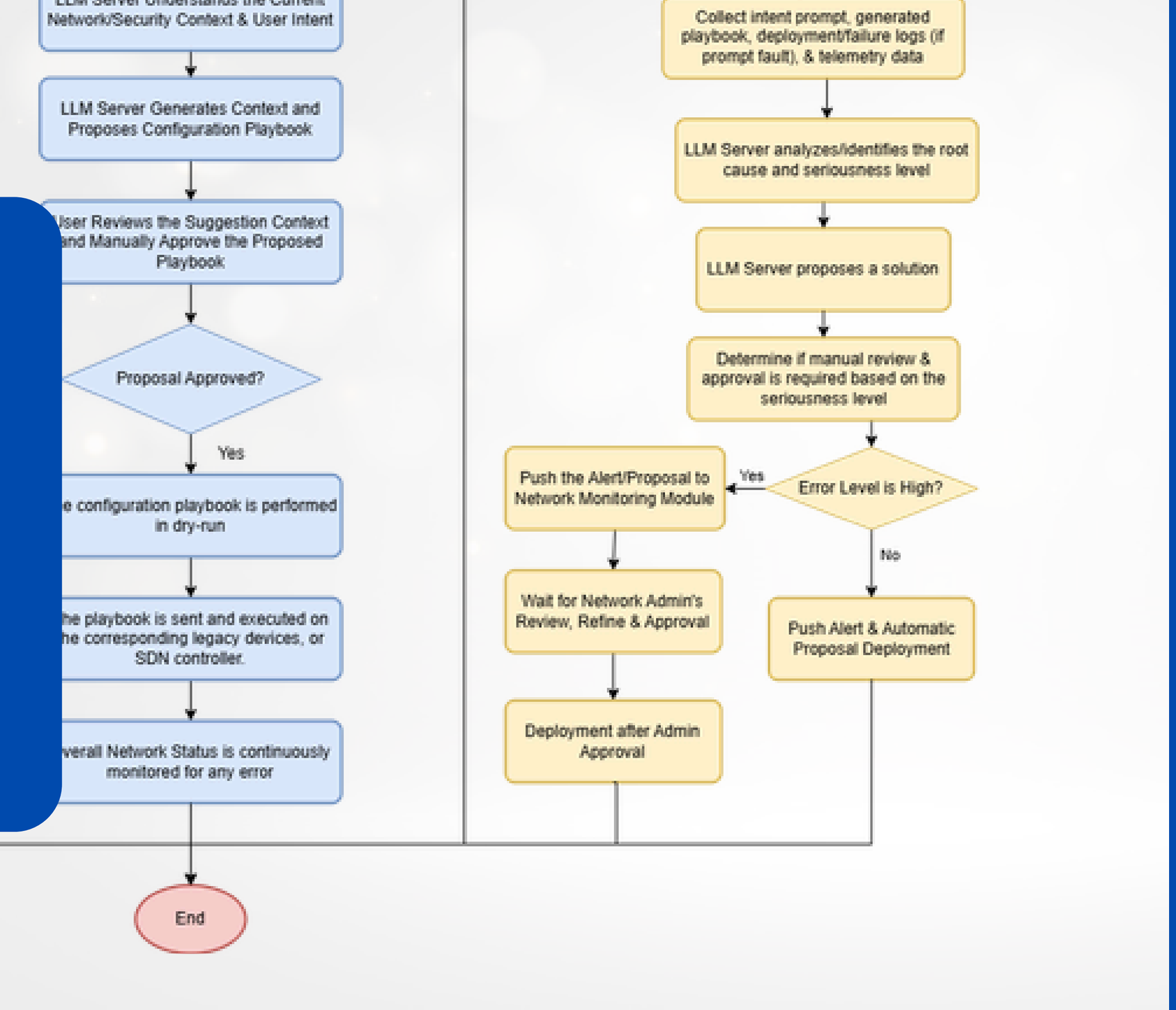
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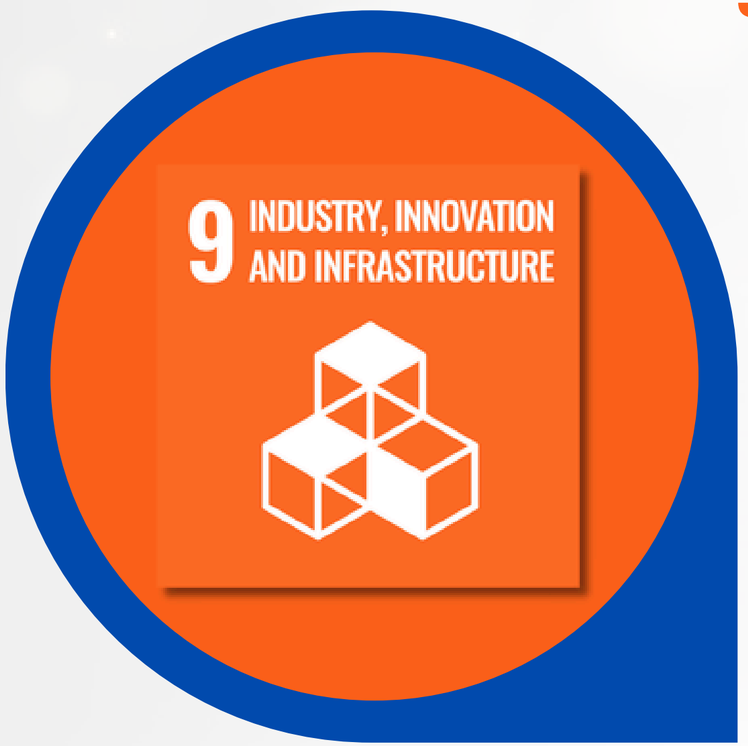


USER ACTIVITY DIAGRAM



USER ACTIVITY DIAGRAM

SUSTAINABLE DEVELOPMENT GOAL



SDG 9 - INDUSTRY, INNOVATION & INFRASTRUCTURE

- **Driving Industry Innovation**

Brings AI and automation to modernize network and security operations.

- **Enabling Open & Inclusive Infrastructure**

Open-source and vendor-neutral for equal access to learning and innovation.

- **Empowering SMEs & Startups**

Cost-effective automation that boosts productivity, scalability, and security.

- **Fostering Community-Driven Development**

Supports collaboration, research, and innovation through an open ecosystem.

CONCLUSION

“**CogniShield** provides a free, accessible, and practical platform for learning and innovation, bringing AI-driven network automation to real networks with structured design, validation, and human oversight.”

Q&A?



THANK YOU !!